

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 4 - LOW**  
**(CODING CHALLENGE)**

**COURSE CODE: ITA04**

**COURSE NAME: STATISTICS WITH R PROGRAMMING**

**COURSE OUTCOME COVERED**

**CO2:** *Implement looping constructs, control structures, and recursive functions in R to solve computational problems with efficient program logic. (BL3, BL4)*

Assessment Tool Weightage: 15%

**QUESTIONS: 20**

**Total Marks: 100**

ANNEXURE A

**SCENARIO BASED QUESTIONS QUESTIONS**

***Code of Conduct:***

*I **TANUJA P (192321052)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

## SCENARIO BASED QUESTIONS QUESTIONS

**1. A data analyst wants to create a function that filters a numeric vector. The function must multiply all positive numbers by 2. If a number is negative or zero, it must remain unchanged. Which control structure is the most efficient and standard way to handle this vectorised operation in R?**

- A) A for loop combined with an if-else statement.
- B) A while loop that increments an index vector.
- C) The vectorised ifelse() function.
- D) A recursive function that processes elements one by one.

**Answer: C) The vectorised ifelse() function.**

---

**2. A developer writes a data cleanup function with the signature `clean_data <- function(df, remove_na = TRUE, threshold = 0.05)`. During a production run, the colleague calls the function using `clean_data(my_df, 0.01)`. What will be the behavior of the arguments inside the function?**

- A) `remove_na` becomes 0.01 and `threshold` uses its default value of 0.05.
- B) R throws a positional argument error because `remove_na` expects a logical value.
- C) `remove_na` retains TRUE and `threshold` is assigned 0.01.
- D) Both arguments are ignored because the positions do not match the data types.

**Answer: A) `remove_na` becomes 0.01 and `threshold` uses its default value of 0.05.**

---

**3. Consider the following R function designed to process a vector:**

```
process_vector <- function(x) {  
  if (any(is.na(x))) {  
    return(NULL)  
  }  
  x * 5  
}
```

**What does this function return if it is called with `process_vector(c(1, 2, 3))`?**

- A) It returns NULL because `return()` was not called at the end of the function.
- B) It returns the vector `c(5, 10, 15)` via implicit return.
- C) It throws a syntax error because a function cannot have both explicit and implicit returns.
- D) It returns nothing (invisible) because the last line lacks an explicit `return()` wrapper.

**Answer: B) It returns the vector `c(5, 10, 15)` via implicit return.**

---

**4. A bioinformatics script needs to return the statistical summaries, the original cleaned data frame, and a diagnostic plot from a single function execution. Since R functions can only return a single object, what is the best way to accomplish this?**

- A) Return a character vector containing the paths to where these objects are saved in the global environment.
- B) Use a pointer to pass the address of the data frame and update it externally.
- C) Combine the data frame, summaries, and plot into a named list object and return the list.
- D) Assign the variables directly to the global environment using the `<<-` operator inside the function without returning anything.

Answer: C) Combine the data frame, summaries, and plot into a named list object and return the list.

---

**5. A Python programmer is learning R and wants to optimize memory usage when working with large matrices. They attempt to pass a matrix to a function to modify its cells directly in-place, expecting pointer-like behavior. Why will this approach fail to mutate the original matrix outside the function scope?**

- A) R requires all matrices to be converted to lists before pointers can reference them.
- B) R passes arguments using a copy-on-write mechanism and does not expose pointers to manipulate memory addresses directly.
- C) The user forgot to activate the memory pointer tracking system using options(pointers = TRUE).
- D) Functions in R can only modify data frames in-place, not matrices.

Answer: B) R passes arguments using a copy-on-write mechanism and does not expose pointers to manipulate memory addresses directly.

---

**6. Review the following snippet of code executing a loop:**

```
val <- 1
while(val < 10) {
  val <- val * 2
  if(val == 4) next
  print(val)
}
```

**What is the exact output printed to the console?**

- A) 2, 4, 8, 16
- B) 2, 8, 16
- C) 2, 4, 8
- D) 2, 8

Answer: B) 2, 8, 16

---

**7. An engineer needs to evaluate whether a patient's lab results meet two conditions simultaneously: the glucose level must be over 100, and the systolic blood pressure must be over 140. They have two vectors representing a population. Which operators should they use to evaluate the entire population at once, ensuring vectorised logical evaluation?**

- A) && and ||
- B) & and |
- C) AND and OR
- D) %in% and ==

Answer: B) & and |

---

**8. In R, "functions are first-class objects." A team wants to leverage this architecture to build a flexible modeling pipeline. Which of the following scenarios best demonstrates this concept?**

- A) Passing a mathematical function like mean or sd as an argument into another function to change summary behaviors dynamically.
- B) Using a for loop to explicitly force a function to run on multiple CPU cores sequentially.
- C) Defining a function inside a spreadsheet cell to calculate column balances automatically.
- D) Ensuring that a function can never be modified or overwritten once it is saved to a package directory.

**Answer: A) Passing a mathematical function like mean or sd as an argument into another function to change summary behaviors dynamically.**

---

**9. A developer writes a recursive function to calculate the nth Fibonacci number:**

```
fib <- function(n) {  
  if (n <= 1) return(n)  
  return(fib(n - 1) + fib(n - 2))  
}
```

**If this function is called with a highly deep nesting level (e.g., fib(50000)), what risk does the developer face in standard R?**

- A) The function will automatically convert the result into a complex matrix, losing precision.
- B) R will trigger a "stack overflow" error because it has a finite expression evaluation limit regulated by options(expressions).
- C) R will bypass the condition and run an infinite loop because recursion forces variables to become pointers.
- D) The function will switch to a vectorised while loop automatically, changing the expected output format.

**Answer: B) R will trigger a "stack overflow" error because it has a finite expression evaluation limit regulated by options(expressions).**

---

**10. A script contains an expression that checks missing values: if (x == NA) { print("Missing") }. When run with a single integer value x <- 5, what will be the result of this conditional statement?**

- A) It will print "Missing" because NA acts as a wildcard match for any integer.
- B) It will evaluate to FALSE and safely skip the print statement.
- C) It will result in an error or warning: argument is of length zero.
- D) It will result in missing value where TRUE/FALSE needed because x == NA outputs NA, not a logical TRUE or FALSE.

**Answer: D) It will result in missing value where TRUE/FALSE needed because x == NA outputs NA, not a logical TRUE or FALSE.**

---

**11. A developer is writing a data intake function and wants a logical argument called verbose to default to FALSE if the user leaves it out. Which function signature achieves this correctly in R?**

- A) read\_data <- function(df, verbose = FALSE)
- B) read\_data <- function(df, verbose <- FALSE)
- C) read\_data <- function(df, verbose(FALSE))
- D) read\_data <- function(df, verbose == FALSE)

**Answer: A) read\_data <- function(df, verbose = FALSE)**

---

**12. You run the following block of code to find the first square number greater than 50:**

```
x <- 1
repeat {
  if (x^2 > 50) {
    print(x^2)
    _____
  }
  x <- x + 1
}
```

**Which keyword must fill the blank to prevent an infinite loop?**

- A) next
- B) return
- C) break
- D) stop

**Answer: C) break**

---

**13. A colleague passes a vector `v <- c(TRUE, FALSE, TRUE)` into an if statement: `if (v) { print("Valid") }`. What will R do when executing this script?**

- A) It evaluates the first element TRUE, prints "Valid", and throws a warning about a condition length > 1.
- B) It automatically applies an implicit all() check and skips the print statement.
- C) It loops through the condition three separate times automatically.
- D) It throws a fatal execution error immediately without reading the block.

**Answer: A) It evaluates the first element TRUE, prints "Valid", and throws a warning about a condition length > 1.**

---

**14. An R script tracks inventory. It must execute specific code when status matches exactly "Low", "Critical", or "Out". Instead of nesting multiple if-else blocks, which native R structure is best suited for matching a single character string against multiple options?**

- A) A nested while loop
- B) The switch() function
- C) A vectorised ifelse() block
- D) The %in% operator inside a repeat loop

**Answer: B) The switch() function**

---

**15. Consider this function that calculates a statistical summary:**

```
analyze <- function(data) {
  res <- summary(data)
  # No return statement here
}
```

**If a user runs `output <- analyze(c(1, 5, 10))`, what will the variable output contain?**

- A) It will contain NULL.
- B) It will contain the summary object evaluated on the last line.
- C) It will be empty, and R will throw an "undefined return value" error.
- D) It will contain a logical TRUE indicating the function finished successfully.

**Answer: B) It will contain the summary object evaluated on the last line.**

---

**16. A programmer attempts to write a recursive function to calculate the factorial of a number:**

```
fact <- function(n) {  
  return(n * fact(n - 1))  
}
```

**What critical component of a recursive function is missing from this snippet, causing an infinite loop?**

- A) The explicit assignment operator
- B) A vectorised loop wrapper
- C) A base case condition to halt execution
- D) A pointer conversion mechanism

**Answer: C) A base case condition to halt execution**

---

**17. An analyst wants to test if a vector x contains at least one negative value. Which logical expression will evaluate the entire vector and return a single TRUE or FALSE value?**

- A)  $x < 0$
- B) `all(x < 0)`
- C) `any(x < 0)`
- D)  $x \ \&\& \ 0$

**Answer: C) `any(x < 0)`**

---

**18. You are reviewing code where a complex matrix calculation happens inside a function. The previous developer ended the function with `return(matrix_result)`. How would removing the explicit `return()` wrapper alter the function's output?**

- A) The function would return NULL instead of the matrix.
- B) The function would print the matrix to the console but return nothing to the variable.
- C) The function behavior would not change; it still returns the matrix implicitly.
- D) R would crash because matrix objects require explicit return handling.

**Answer: C) The function behavior would not change; it still returns the matrix implicitly.**

---

**19. A data pipeline uses the expression `result <- A & B`. If A is `c(TRUE, FALSE)` and B is `c(TRUE, TRUE)`, what is the value of result?**

- A) TRUE
- B) `c(TRUE, FALSE)`
- C) `c(TRUE, TRUE)`
- D) FALSE

**Answer: B) `c(TRUE, FALSE)`**

---

**20. An engineer writes a function that needs to return both a vector of coefficients and a diagnostic data frame. They want to make sure the user can easily extract these components by name using the \$ operator. What object should the function return?**

- A) A matrix
- B) A multi-dimensional array
- C) A named list
- D) An unlisted character vector

**Answer: C) A named list**

---

**21. Because functions in R are first-class objects, you can store them inside a list.**

**Review the following snippet:**

```
math_ops <- list(  
  add_one = function(x) x + 1,  
  double = function(x) x * 2  
)
```

**How do you invoke the second function in this list to process the number 5?**

- A) `math_ops$double(5)`
- B) `math_ops[] * 5`
- C) `call(math_ops, double, 5)`
- D) `math_ops(2, 5)`

**Answer: A) `math_ops$double(5)`**

---

**22. A script defines a simple loop:**

```
for (i in 1:3) {  
  if (i == 2) next  
  print(i)  
}
```

**What is the exact output displayed in the console?**

- A) 1, 2, 3
- B) 1, 3
- C) 1
- D) 3

**Answer: B) 1, 3**

---

**23. A user wants to compare two numeric values x and y while completely avoiding short-circuiting behavior. They want to ensure both sides of the logical expression are evaluated every time. Which operator should they choose?**

- A) `&&`
- B) `&`
- C) `||`
- D) `and`

**Answer: B) `&`**

---

**24. A data cleaning function relies on a default threshold: `process <- function(val, cut = 10)`. If a developer calls this function using `process(5, cut = NULL)`, what value will the argument `cut` hold during execution?**

- A) It falls back to its default value of 10.
- B) It evaluates to `NULL`, overriding the default value.
- C) R throws an error because `NULL` cannot be assigned to an argument with a default.
- D) It evaluates to `NA`.

**Answer: B) It evaluates to `NULL`, overriding the default value.**

---

**25. An engineer coming from a C++ background wants to speed up an R script by passing a data frame pointer to a function. They want the function to add a column directly to the original global data frame without copying it. Why is this unachievable in standard R?**

- A) Data frames can only be modified inside a while loop, not inside custom functions.
- B) R does not feature user-exposed pointer variables; object adjustments rely on copy-on-write or explicit reassignment.
- C) Global variables are completely locked and invisible inside custom functions.
- D) R functions can only return primitive atomic vectors, not structured tables.

**Answer: B) R does not feature user-exposed pointer variables; object adjustments rely on copy-on-write or explicit reassignment.**

---

**26. What is the value of the logical expression `5 > 3 | 10 < 2` in R?**

- A) `TRUE`
- B) `FALSE`
- C) `NA`
- D) `NULL`

**Answer: A) `TRUE`**

---

**27. Examine the following loop construct:**

```
count <- 1
while(count < 4) {
  count <- count + 1
}
print(count)
```

**What value is printed to the console when the loop finishes?**

- A) 3
- B) 4
- C) 5
- D) 2

**Answer: B) 4**

---

**28. A function needs to terminate early and return a specific message if an input vector is empty. In which scenario is using an explicit `return()` statement highly recommended over an implicit return?**



- A) At the very last line of the function block.
- B) Inside a conditional guard clause at the beginning of the function to exit early.
- C) Inside a standard variable assignment block.
- D) R functions never support explicit returns; they are always implicit.

**Answer: B) Inside a conditional guard clause at the beginning of the function to exit early.**

---

**29. A script evaluates an expression: `res <- TRUE && FALSE`. What is the value of `res`?**

- A) TRUE
- B) FALSE
- C) `c(TRUE, FALSE)`
- D) NA

**Answer: B) FALSE**

---

**30. A team is building an optimization loop. They want a loop structure that is guaranteed to run the code block at least once before checking the conditional exit criteria. Which structure fits this requirement best?**

- A) A standard for loop
- B) A while loop
- C) A repeat loop containing a conditional break at the end
- D) A recursive function call

**Answer: C) A repeat loop containing a conditional break at the end**

---

**31. A script runs the following command: `val <- (TRUE | NA)`. What is the value of `val`?**

- A) NA
- B) FALSE
- C) TRUE
- D) Error: logical operator cannot handle missing values

**Answer: C) TRUE**

---

**32. A package developer writes a function where one argument depends on another:**  
`scale_data <- function(x, center = mean(x)) {`

```
  x - center
}
```

**If a user runs `scale_data(c(10, 20, 30))`, how does R evaluate the default argument `center`?**

- A) R throws an error because `x` is not yet defined when the function signature is parsed.
- B) R uses lazy evaluation, computing `mean(x)` only when `center` is explicitly referenced inside the function body.
- C) R sets `center` to `NULL` automatically because it cannot look ahead.
- D) R searches the global environment for a variable named `x` before checking the function input.

**Answer: B) R uses lazy evaluation, computing mean(x) only when center is explicitly referenced inside the function body.**

---

**33. Which arithmetic operator is used in R to calculate the remainder of a division (modulo operation)?**

- A) %/%%
- B) mod
- C) %
- D) %%%

**Answer: D) %%%**

---

**34. What happens if you try to assign a function to a new variable name, such as my\_log <- log?**

- A) R executes the function immediately and saves the numerical result.
- B) R throws a duplicate assignment syntax error.
- C) my\_log becomes an alias for the log function object and can be called using my\_log(10).
- D) The original log function is deleted from system memory.

**Answer: C) my\_log becomes an alias for the log function object and can be called using my\_log(10).**

---

**35. You want to execute a block of code only if a score is between 80 and 90 inclusive. Which conditional statement is written correctly for a single scalar value?**

- A) if (80 <= score <= 90)
- B) if (score >= 80 & score <= 90)
- C) if (score >= 80 && score <= 90)
- D) if (score >= 80 AND score <= 90)

**Answer: C) if (score >= 80 && score <= 90)**

---

**36. An analyst wants to write a function that calculates the weighted average of a vector. If the user does not provide weights, the function should automatically default to calculating the standard unweighted arithmetic mean. Which function definition correctly implements this option?**

- A) calculate\_average <- function(x, weights = rep(1, length(x)))
- B) calculate\_average <- function(x, weights <- rep(1, length(x)))
- C) calculate\_average <- function(x, weights == mean(x))
- D) calculate\_average <- function(x, weights = default)

**Answer: A) calculate\_average <- function(x, weights = rep(1, length(x)))**

---

**37. Review the following code snippet evaluating a logical state:**

```
x <- c(10, 20, 30)
y <- c(5, 25, 15)
res <- (x > 15) && (y > 20)
```

**What is the final exact value stored inside the scalar variable res?**

- A) FALSE
- B) TRUE

C) `c(FALSE, TRUE, FALSE)`

D) `NA`

**Answer: A) FALSE**

---

**38. A researcher is building a structural validation function that tests clean dataset dimensions. Under what specific architectural circumstance must the developer explicitly invoke the `return()` function?**

A) When outputting a multi-dimensional array or a nested list structure at the end of the script.

B) Inside an early-exit conditional block to interrupt function execution before hitting the terminal line.

C) When the output object contains missing numeric values or logical `NA` elements.

D) R functions strictly demand explicit returns for complex matrices and never support implicit output structures.

**Answer: B) Inside an early-exit conditional block to interrupt function execution before hitting the terminal line.**

---

**39. A system engineer is migrating code from an environment where functions are strictly declarative subroutines to R. They want to store three different metrics models inside an R object and pass that entire container into a processing engine. Why does R allow functions to be handled as elements inside data collections?**

A) Because R converts all functional blocks into text strings before executing them.

B) Because R lacks pointers and treats code statements as raw data tables instead.

C) Because functions in R are first-class objects, meaning they can be treated like any other variable or value type.

D) Because functions are automatically mapped to global memory slots when defined.

**Answer: C) Because functions in R are first-class objects, meaning they can be treated like any other variable or value type.**

---

**40. Examine the following tracking loop designed to locate a specific threshold condition:**

```
val <- 2
while (val < 7) {
  val <- val + 2
  if (val == 6) break
}
print(val)
```

**What exact final value is printed to the R console when the control structure exits?**

A) 4

B) 6

C) 8

D) 2

**Answer: B) 6**

## RUBRICS FOR CALCULATION

| Criteria                                 | Weightage | Level 4 – Exemplary                                                                  | Level 3 – Proficient                                  | Level 2 – Developing                                      | Level 1 – Beginning                                     |
|------------------------------------------|-----------|--------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------|
| Problem Understanding & Context Analysis | 25        | Thoroughly understands the scenario with accurate interpretation of all requirements | Clearly understands the scenario with minor gaps      | Basic understanding; some aspects of the scenario unclear | Poor understanding or misinterpretation of the scenario |
| Application of Concepts                  | 30        | Applies appropriate concepts effectively to solve complex real-world problems        | Applies relevant concepts correctly with minor errors | Applies basic concepts but with limited accuracy          | Fails to apply appropriate concepts                     |
| Analytical & Decision-Making Skills      | 20        | Demonstrates strong analysis and selects optimal solutions with justification        | Good analysis with reasonable solutions               | Limited analysis; solutions lack justification            | Poor or no analysis; incorrect decisions                |

|                                       |    |                                                             |                                                 |                                                      |                                        |
|---------------------------------------|----|-------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------|----------------------------------------|
| Solution Design & Approach            | 15 | Well-structured, innovative, and feasible solution approach | Logical and structured solution with minor gaps | Basic solution with limited structure or feasibility | Unclear or incorrect solution approach |
| Clarity, Justification & Presentation | 10 | Clear, well-justified explanation with strong reasoning     | Clear explanation with some justification       | Limited clarity and weak justification               | Poorly explained with no justification |